

Digital Electronics Question and Answers

Logic Families

Topic 3

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1.) Register-Transistor Logic(RTL)

1. All input of NOR as low produces result as _____

- a) Low
- b) Mid
- c) High
- d) Floating

Answer: c

Explanation: All input of NOR as low produces the result as high, whereas, rest all conditions produce output as low.

2. In RTL NOR gate, the output is at logic 1 only when all the inputs are at _____

- a) logic 0
- b) logic 1
- c) +10V
- d) Floating

Answer: a

Explanation: RTL NOR gate behaves as NOR gate and the output of NOR gate will be 1 only when all the inputs are at logic 0 and in rest conditions of the input, the output is 0.

3. Resistor–transistor logic (RTL) is a class of digital circuits built using _____ as the input network and _____ as switching devices.

- a) Resistors, bipolar junction transistors (BJTs)
- b) Bipolar junction transistors (BJTs), Resistors
- c) Capacitors, resistors
- d) Resistors, capacitors

Answer: a

Explanation: Resistor–transistor logic (RTL) is a class of digital circuits built using resistors as the input network and bipolar junction transistors (BJTs) as switching devices.

4. RTL consists of a common emitter stage with a _____ connected between the base and the input voltage source.

- a) collector
- b) base resistor

- c) capacitor
- d) inductor

Answer: b

Explanation: RTL consist of a common emitter stage with a base resistor connected between the base and the input voltage source. The role of base resistor is to expand the negligible transistor input voltage range (about 0.7 V) to the logical "1" level (about 3.5 V) by converting the input voltage into a current. Thus, base resistor plays a major role in biasing of the transistor.

5. The role of the _____ is to convert the collector current into a voltage in RTL.

- a) Collector resistor
- b) Base resistor
- c) Capacitor
- d) Inductor

Answer: a

Explanation: The role of the collector resistor is to convert the collector current into a voltage; its resistance is chosen high enough to saturate the transistor and low enough to obtain low output resistance. Base Resistor is to provide the necessary biasing to the base of the transistor in order to activate it.

6. The limitations of the one transistor RTL NOR gate are overcome by _____

- a) Two-transistor RTL implementation
- b) Three-transistor RTL implementation
- c) Multi-transistor RTL implementation
- d) Four-transistor RTL implementation

Answer: c

Explanation: The limitations of the one transistor RTL NOR gate are overcome by the use of multi transistor RTL. It consists of a set of parallel connected transistor switches driven by the logic inputs.

7. The primary advantage of RTL technology was that _____

- a) It results as low power dissipation
- b) It uses a minimum number of resistors
- c) It uses a minimum number of transistors
- d) It operates swiftly

Answer: c

Explanation: The primary advantage of RTL technology was that it uses a minimum number of transistors. It consists of registers in large amount and it results in as high

power dissipation. The resistors act as the input network and the transistors performs the switching operation.

8. The disadvantage of RTL is that _____

- a) It uses a maximum number of resistors
- b) It results in high power dissipation
- c) High noise creation
- d) It uses a minimum number of transistors

Answer: b

Explanation: The disadvantage of RTL is its high power dissipation when the transistor is switched on by current flowing in the collector and base resistor. This requires that more current be supplied to and heat be removed from RTL circuits. In contrast, TTL circuits with “totem-pole” output stage minimize both of these requirements.

9. TTL circuits with “totem-pole” output stage minimize _____

- a) The power dissipation in RTL
- b) The time consumption in RTL
- c) The speed of transferring rate in RTL
- d) Propagation delay in RTL

Answer: a

Explanation: TTL circuits with “totem-pole” output stage minimize the power dissipation and heating effect in RTL.

10. The minimum number of transistors can be used by 2 input AND gate is _____

- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: The minimum number of transistors can be used by 2 input AND gate is 2 and maximum up to 3.

2.) Diode-Transistor Logic(DTL)

1. Diode–transistor logic (DTL) is the direct ancestor of _____

- a) Register-transistor logic
- b) Transistor–transistor logic
- c) High threshold logic
- d) Emitter Coupled Logic

Answer: b

Explanation: Diode–transistor logic (DTL) is a class of digital circuits that is the direct ancestor of transistor–transistor logic. To overcome the shortcomings of DTL, TTL came into existence.

2. In DTL logic gating function is performed by _____

- a) Diode
- b) Transistor
- c) Inductor
- d) Capacitor

Answer: a

Explanation: Diode serves as the input network and the switching operation is performed by the transistor.

3. In DTL amplifying function is performed by _____

- a) Diode
- b) Transistor
- c) Inductor
- d) Capacitor

Answer: b

Explanation: The amplifying and switching function is performed by a transistor and the diode acts as an input network in DTL.

4. How many stages a DTL consist of?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: The DTL circuit shown in the picture consists of three stages: an input diode logic stage, an intermediate level shifting stage and an output common-emitter amplifier stage.

5. The full form of CTDL is _____

- a) Complemented transistor diode logic
- b) Complemented transistor direct logic
- c) Complementary transistor diode logic
- d) Complementary transistor direct logic

Answer: a

Explanation: The full form of CTDL is Complemented transistor diode logic.

6. The DTL propagation delay is relatively _____

- a) Large
- b) Small
- c) Moderate
- d) Negligible

Answer: a

Explanation: Propagation delay refers to the time taken by the output to change its state when the input is altered. When the transistor goes into saturation from all inputs being high charge is stored in the base region. When it comes out of saturation (one input goes low) this charge has to be removed and will dominate the propagation time which results as a large propagation delay. Thus, it has small clock frequency.

7. The way to speed up DTL is to add an across intermediate resistor is _____

- a) Small "speed-up" capacitor
- b) Large "speed-up" capacitor
- c) Small "speed-up" transistor
- d) Large "speed-up" transistor

Answer: a

Explanation: One way to speed up DTL is to add a small "speed-up" capacitor across intermediate resistor. The capacitor helps to turn off the transistor by removing the stored base charge; the capacitor also helps to turn on the transistor by increasing the initial base drive.

8. The process to avoid saturating the switching transistor is performed by _____

- a) Baker clamp
- b) James R. Biard
- c) Chris Brown
- d) Totem-Pole

Answer: a

Explanation: Another way to speed up DTL other than adding a small "speed-up" capacitor across intermediate resistor is to avoid saturating the switching transistor which can be done with a Baker clamp. The name Baker clamp is given at the name of Richard H. Baker, who described it in his 1956 technical report "Maximum Efficiency Switching Circuits".

9. A major advantage of DTL over the earlier resistor–transistor logic is the _____

- a) Increased fan out
- b) Increased fan in
- c) Decreased fan out
- d) Decreased fan in

Answer: b

Explanation: A major advantage over the earlier resistor–transistor logic is the increased fan in. Fan-in is the measure of the maximum number of inputs that a single gate output can accept.

10. To increase fan-out of the gate in DTL _____

- a) An additional capacitor may be used
- b) An additional resistor may be used
- c) An additional transistor and diode may be used
- d) Only an additional diode may be used

Answer: c

Explanation: To increase fan-out of the gate in DTL, an additional transistor and diode may be used. Here, the fan out means the number of maximum input that a single gate output can feed.

11. A disadvantage of DTL is _____

- a) The input transistor to the resistor
- b) The input resistor to the transistor
- c) The increased fan-in
- d) The increased fan-out

Answer: b

Explanation: A disadvantage of DTL is the input resistor to the transistor and its presence tends to slow the circuit down. Hence limiting the speed at which the transistor is able to switch states. Thus, the propagation delay increases.

3.) Transistor Transistor Logic(TTL)

1. Transistor–transistor logic (TTL) is a class of digital circuits built from _____

- a) JFET only
- b) Bipolar junction transistors (BJT)
- c) Resistors
- d) Bipolar junction transistors (BJT) and resistors

Answer: d

Explanation: Transistor–transistor logic (TTL) is a class of digital circuits built from bipolar junction transistors (BJT) and resistors. However, resistors have a small role to play and both logic gating and amplifying functions are performed by the transistors.

2. TTL is called transistor–transistor logic because both the logic gating function and the amplifying function are performed by _____

- a) Resistors
- b) Bipolar junction transistors
- c) One transistor
- d) Resistors and transistors respectively

Answer: b

Explanation: TTL is called transistor–transistor logic because both the logic gating function and the amplifying function are performed by bipolar junction transistors (BJTs).

3. TTL was invented in 1961 by _____

- a) Baker clamp
- b) James L. Buie
- c) Chris Brown
- d) Frank Wanlass

Answer: b

Explanation: TTL was invented in 1961 by James L Buie.

4. The full form of TCTL is _____

- a) Transistor-coupled transistor logic
- b) Transistor-capacitor transistor logic
- c) Transistor-complemented transistor logic
- d) Transistor-complementary transistor logic

Answer: a

Explanation: The full form of TCTL is transistor-coupled transistor logic.

5. The _____ ancestor to the first personal computers.

- a) PARAM 1
- b) SATYAM 1
- c) KENBAK 1
- d) MITS Altair

Answer: c

Explanation: The KENBAK 1, ancestor to the first personal computers.

6. TTL inputs are the emitters of a _____

- a) Transistor-transistor logic
- b) Multiple-emitter transistor
- c) Resistor-transistor logic
- d) Diode-transistor logic

Answer: b

Explanation: TTL inputs are the emitters of a multiple-emitter transistor.

7. TTL is a _____

- a) Current sinking
- b) Current sourcing
- c) Voltage sinking
- d) Voltage sourcing

Answer: a

Explanation: Like DTL, TTL is a current-sinking logic since a current must be drawn from inputs to bring them to a logic 0 level. Current Sink means it accepts current coming out from a source.

8. Standard TTL circuits operate with a ___ volt power supply.

- a) 2
- b) 4
- c) 5
- d) 3

Answer: c

Explanation: Standard TTL circuits operate with a 5-volt power supply.

9. TTL devices consume substantially _____ power than equivalent CMOS devices at rest.

- a) Less
- b) More
- c) Equal
- d) Very High

Answer: b

Explanation: TTL devices consume substantially more power than equivalent CMOS devices at rest. Thus, CMOS devices are faster than TTL devices.

10. A TTL gate may operate inadvertently as an _____

- a) Digital amplifier
- b) Analog amplifier

- c) Inverter
- d) Regulator

Answer: b

Explanation: A TTL gate may operate inadvertently as an analog amplifier if the input is connected to a slowly changing input signal that traverses the unspecified region from 0.7V to 3.3V.

11. The speed of _____ circuits is limited by the tendency of common emitter circuits to go into saturation.

- a) TTL
- b) ECL
- c) RTL
- d) DTL

Answer: a

Explanation: The speed of TTL circuits is limited by the tendency of common emitter circuits to go into saturation due to the injection of minority carriers into the collector region. Hence, it functions slowly compared to CMOS devices.

4.) Emitter-Coupled Logic(ECL)

1. The full form of ECL is _____

- a) Emitter-collector logic
- b) Emitter-complementary logic
- c) Emitter-coupled logic
- d) Emitter-cored logic

Answer: c

Explanation: The full form of ECL is emitter-coupled logic.

2. Which logic is the fastest of all the logic families?

- a) TTL
- b) ECL
- c) HTL
- d) DTL

Answer: b

Explanation: ECL is the fastest of all the logic families because of the emitters of many transistors are coupled together which results in the highest transmission rate.

3. The full form of CML is _____

- a) Complementary mode logic

- b) Current mode logic
- c) Collector mode logic
- d) Collector Mixed Logic

Answer: c

Explanation: The full form of CML is Collector Mode Logic.

4. Sometimes ECL can also be named as _____

- a) EEL
- b) CEL
- c) CML
- d) CCL

Answer: c

Explanation: ECL (Emitter Coupled Logic) can also be named as CML(Collector Mode Logic).

5. In an ECL the output is taken from _____

- a) Emitter
- b) Base
- c) Collector
- d) Junction of emitter and base

Answer: c

Explanation: Though, the emitter and collector of the ECL are coupled together. So, the output will be taken from a collector.

6. The ECL behaves as _____

- a) NOT gate
- b) NOR gate
- c) NAND gate
- d) AND gate

Answer: b

Explanation: The ECL behaves as NOR gate because if any of the input voltages go high as compared to the reference voltage, the output is low and the output is high only when all the input voltages are low.

7. In ECL the fanout capability is _____

- a) High
- b) Low
- c) Zero
- d) Sometimes high and sometimes low

Answer: a

Explanation: If the input impedance is high and the output resistance is low; as a result, the transistors change states quickly, gate delays are low, and the fanout capability is high. Fan-out is the measure of the maximum number of inputs that a single gate output can accept.

8. ECL's major disadvantage is that _____

- a) It requires more power
- b) It's fanout capability is high
- c) It creates more noise
- d) It is slow

Answer: a

Explanation: ECL's major disadvantage is that each gate continuously draws current, which means it requires (and dissipates) significantly more power than those of other logic families. But ECL logic gates have clock frequency. Thus, they have a fast operation.

9. The full form of SCFL is _____

- a) Source-collector logic
- b) Source-coupled logic
- c) Source-complementary logic
- d) Source Cored Logic

Answer: b

Explanation: The full form of SCFL is source-coupled logic.

10. The equivalent of emitter-coupled logic made out of FETs is called _____

- a) CML
- b) SCFL
- c) FECL
- d) EFCL

Answer: b

Explanation: The equivalent of emitter-coupled logic made out of FETs is called Source-coupled logic(SCFL). Like ECL, SCL is also the fastest among the logic families.

11. ECL was invented in _____ by _____

- a) 1956, Baker clamp
- b) 1976, James R. Biard
- c) 1956, Hannon S. Yourke
- d) 1976, Yourke

Answer: c

Explanation: ECL was invented in August 1956 at IBM by Hannon S Yourke.

12. At the time of invention, an ECL was called as _____

- a) Source-coupled logic
- b) Current Mode Logic
- c) Current-steering logic
- d) Emitter-coupled logic

Answer: c

Explanation: At the time of invention, an ECL was called a current-steering logic because it involved current switching.

13. The ECL circuits usually operates with _____

- a) Negative voltage
- b) Positive voltage
- c) Grounded voltage
- d) High Voltage

Answer: a

Explanation: The ECL circuits usually operate with negative power supplies (positive end of the supply is connected to ground), in comparison to other logic families in which negative end of the supply is grounded. It is done mainly to minimize the influence of the power supply variations on the logic levels as ECL is more sensitive to noise on the VCC and relatively immune to noise on VEE.

14. Low-voltage positive emitter-coupled logic (LVPECL) is a power optimized version of _____

- a) ECL
- b) VECL
- c) PECL
- d) LECL

Answer: c

Explanation: Low voltage positive emitter coupled logic (LVPECL) is a power optimized version of PECL using a +3.3 V instead of 5 V supply.

5.) Integrated-injection Logic(I²L)

1. The full form of IIL is _____

- a) Injection integrated logic
- b) Integrated inspected logic

- c) Integrated injection logic
- d) Injection Inspected Logic

Answer: c

Explanation: The full form of IIL is Integrated injection logic. It is made up of multiple bipolar junction transistors and had a speed comparable to TTL and power consumption equivalent to CMOS.

2. Integrated injection logic is a class of digital circuits built with _____

- a) Single collector BJT
- b) Double emitter BJT
- c) Multiple emitter BJT
- d) Multiple collector BJT

Answer: d

Explanation: Integrated injection logic is a class of digital circuits built with multiple collector bipolar junction transistors (BJT).

3. IIL has _____ noise immunity.

- a) High
- b) Low
- c) Neutral
- d) Nil

Answer: a

Explanation: The full form of IIL is Integrated injection logic. It is made up of multiple bipolar junction transistors and had speed comparable to TTL and power consumption equivalent to CMOS. IIL has high noise immunity because it operates by current instead of voltage.

4. IIL is sometimes also known as _____

- a) Single transistor logic
- b) Multiple transistor logic
- c) Merged transistor logic
- d) Mixed Transistor Logic

Answer: c

Explanation: The full form of IIL is Integrated injection logic. It is made up of multiple bipolar junction transistors and had speed comparable to TTL and power consumption equivalent to CMOS. IIL is sometimes also known as merged transistor logic.

5. Integrated Injection logic cannot be denoted as _____

- a) IIL

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- b) I²L
- c) I²L
- d) I TWO L

Answer: d

Explanation: The full form of IIL is Integrated injection logic. It is made up of multiple bipolar junction transistors and had speed comparable to TTL and power consumption equivalent to CMOS. Integrated Injection logic can be denoted as IIL, I²L and I²L as well.

6. The heart of an I²L circuit is the _____

- a) Common collector open emitter inverter
- b) Common emitter open collector inverter
- c) Open emitter common collector inverter
- d) Open Emitter Open Collector Inverter

Answer: b

Explanation: The heart of an I²L circuit is the common emitter open collector inverter. The emitter is connected to the ground, with the base having an input behaving as either current sink or high impedance floating condition. The output of an inverter is connected to the collector.

7. In Integrated Injection, logic input is supplied to the _____

- a) Emitter
- b) Base
- c) Collector
- d) Collector Emitter

Answer: b

Explanation: In IIL, the input is supplied to the base as either a current sink (low logic level) or as a high impedance floating condition (high logic level). The output of an inverter is connected to the collector.

8. In Integrated Injection, logic output is received through the _____

- a) Emitter
- b) Base
- c) Collector
- d) Base Emitter Junction

Answer: c

Explanation: In Integrated Injection logic output is received through the collector. The output of an inverter is connected at the collector.

9. When the outputs of two inverters are wired together the result is _____

- a) A two-input NOR gate
- b) A single-input NOR gate
- c) A two-input NAND gate
- d) A single-input NAND gate

Answer: a

Explanation: When the outputs of two inverters are wired together then the result is a two-input NOR gate because the configuration (NOT A) AND (NOT B) is equivalent to NOT (A OR B). In a NOR gate, the output is high only when all of its input are at logic 0 else, the output is low.

10. I²L was commonly used before the advent of CMOS logic by company _____

- a) Samsung
- b) Nokia
- c) Motorola
- d) Apple

Answer: c

Explanation: I²L is relatively simple to construct on an integrated circuit and was commonly used before the advent of CMOS logic by companies such as Motorola.

11. Which company introduced one of the first consumer-grade digital watches (the Black Watch) which used I²L technology?

- a) Cambridge Consultants Pvt Ltd
- b) Sinclair Radionics Ltd
- c) National Research and Development Council
- d) Westminster

Answer: b

Explanation: Sinclair Radionics Ltd introduced one of the first consumer-grade digital watches (the Black Watch) in August 1975, which used I²L technology.

6.) MOS Digital Integrated Circuits

1. The full form of MOS is _____

- a) Metal oxide semiconductor
- b) Metal oxygen semiconductor
- c) Metallic oxide semiconductor
- d) Metallic oxygen semiconductor

Answer: a

Explanation: The full form of MOS is “Metal Oxide Semiconductor”. It is a type of

transistor having 3 layers: a metal conductor, an insulating silicon layer and a semiconductor silicon layer.

2. What are the types of MOSFET devices available?

- a) P-type enhancement type MOSFET
- b) N-type enhancement type MOSFET
- c) Depletion type MOSFET
- d) All of the mentioned

Answer: d

Explanation: MOSFET are of two types: enhancement and depletion type. Further, these are classified into n-type and p-type device. The depletion type is switched on without the application of gate bias voltage and the enhancement type is switched on with the application of gate voltage.

3. Which insulating layer used in the fabrication of MOSFET?

- a) Aluminium oxide
- b) Silicon Nitride
- c) Silicon dioxide
- d) Aluminium Nitrate

Answer: c

Explanation: Silicon dioxide is used as an insulating layer in the fabrication of MOSFET. It gives an extremely high input resistance in the order of 10^{10} to $10^{15} \Omega$ for MOSFET.

4. Which of the following plays an important role in improving the device performance of MOSFET?

- a) Dielectric constant
- b) Threshold voltage
- c) Power supply voltage
- d) Gate to drain voltage

Answer: b

Explanation: In MOSFET, the threshold voltage is typically 3 to 6V. This large voltage is not compatible with the supply of 5V which is used in digital ICs. So, for the improvement of the device's performance the magnitude of threshold voltage should be reduced.

5. A technique used to reduce the magnitude of threshold voltage of MOSFET is the

-
- a) Use of complementary MOSFET
 - b) Use of Silicon nitride

- c) Using thin film technology
- d) Increasing potential of the channel

Answer: b

Explanation: Silicon nitride is sandwiched between two SiO₂ layer and provide necessary barrier. The dielectric constant of Si₃N₄ is 7.5, whereas that of SiO₂ is 4. This increase in overall dielectric constant reduces threshold voltage.

6. What is used to higher the speed of operation in MOSFET fabrication?

- a) Ceramic gate
- b) Silicon dioxide
- c) Silicon nitride
- d) Poly silicon gate

Answer: d

Explanation: In conventional metal gate small overlap capacitance is present, which lowers the speed of operation. With the presence of self aligning property of the poly silicon gate it eliminates this capacitance. Using a process called ion-implantation, polysilicon, the drain and the source get doped. However, the thin oxide under silicon gate acting as a mask for the process and thus develops the gate aligning property.

7. Find the sequence of steps involved in fabrication of poly silicon gate MOSFET?

Step 1: Entire wafer surface of a Si₃N₄ is coated and is etched away with the help of mask to include

source, gate and drain.

Step 2: The contact areas are defined using photolithographic process.

Step 3: Selective etching of Si₃N₄ and growth of thin oxide.

Step 4: The deposition of poly silicon gate.

Step 5: The growth of thick oxide is called field oxide and P implantation.

Step 6: The metallization and interconnection between substrate and source.

- a) 1->5->3->4->2->6
- b) 1->3->4->2->5->6
- c) 1->5->4->3->2->6
- d) 1->4->2->5->3->6

Answer: a

Explanation: These steps are the sequence of steps involved in fabrication of poly silicon

gate MOSFET. With the help of poly silicon gate doping, it highers the speed of operation of the MOSFET.

8. Why MOSFET is preferred over BJT in IC components?

- a) MOSFET has low packing density
- b) MOSFET has medium packing density
- c) MOSFET has high packing density
- d) MOSFET has no packing density

Answer: a

Explanation: MOSFET is preferred over BJT because of its low packaging density. Thus, more number of MOSFET memory cells can be accommodated in a particular area as compared to BJT.

9. Critical defects per unit chip area are _____ for a MOS transistor.

- a) High
- b) Low
- c) Neutral
- d) Very High

Answer: b

Explanation: Critical defects per unit chip area is low for a MOS transistor because it involves fewer steps in the fabrication of a MOS transistor. Also, MOSFET has low packaging density.

10. MOS is being used in _____

- a) LSI
- b) VLSI
- c) MSI
- d) Both LSI and VLSI

Answer: d

Explanation: Since more transistor and circuitry functions can be achieved on a single chip with MOS technology that is why MOS is being used in LSI and VLSI. LSI stands for Large Scale Integration and VLSI stands for Very Large Scale Integration.

7.) Characteristics of CMOS

1. The full form of CMOS is _____

- a) Capacitive metal oxide semiconductor
- b) Capacitive metallic oxide semiconductor

- c) Complementary metal oxide semiconductor
- d) Complemented metal oxide semiconductor

Answer: c

Explanation: The full form of CMOS is complementary metal oxide semiconductor. In this type of device, both n-type and p-type transistors are used in a complementary way.

2. The full form of COS-MOS is _____
- a) Complementary symmetry metal oxide semiconductor
 - b) Complementary systematic metal oxide semiconductor
 - c) Capacitive symmetry metal oxide semiconductor
 - d) Complemented systematic metal oxide semiconductor

Answer: a

Explanation: The full form of COS-MOS is complementary systematic metal oxide semiconductor. In this type of device, both n-type and p-type transistors are used in a complementary way. Usually, the transistors used are MOSFETs.

3. CMOS is also sometimes referred to as _____
- a) Capacitive metal oxide semiconductor
 - b) Capacitive symmetry metal oxide semiconductor
 - c) Complementary symmetry metal oxide semiconductor
 - d) Complemented symmetry metal oxide semiconductor

Answer: c

Explanation: CMOS is also sometimes referred to as complementary systematic metal oxide–semiconductor (COS-MOS). In this type of device, both n-type and p-type transistors are used in a complementary way. Usually, the transistors used are MOSFETs.

4. CMOS technology is used in _____
- a) Inverter
 - b) Microprocessor
 - c) Digital logic
 - d) Both microprocessor and digital logic

Answer: d

Explanation: CMOS technology is used in Microprocessor, Microcontroller, static RAM and other digital logic circuits. CMOS technology is also used for several analog circuits such as image sensors (CMOS sensor), data converters and highly integrated transceivers for many types of communication.

5. Two important characteristics of CMOS devices are _____

- a) High noise immunity
- b) Low static power consumption
- c) High resistivity
- d) Both high noise immunity and low static power consumption

Answer: d

Explanation: Two important characteristics of CMOS devices are high noise immunity and low static power consumption. Since one transistor of the pair is always off and the series combination draws significant power only momentarily during switching between on and off states. Also, the performance of CMOS is not altered with the presence of noise and thus it has high noise immunity.

6. CMOS behaves as a/an _____

- a) Adder
- b) Subtractor
- c) Inverter
- d) Comparator

Answer: c

Explanation: Since, the outputs of the PMOS and NMOS transistors are complementary such that when the input is low, the output is high and when the input is high, the output is low. Because of this behaviour of input and output, the CMOS circuit's output is the inverse of the input. Whereas, adders and subtractors are combinational circuits.

7. An important characteristic of a CMOS circuit is the _____

- a) Noise immunity
- b) Duality
- c) Symmetricity
- d) Noise Margin

Answer: b

Explanation: An important characteristic of a CMOS circuit is the duality that exists between its PMOS transistors and NMOS transistors. Due to the presence of two different types of transistors, the device has a complementary function.

8. CMOS logic dissipates _____ power than NMOS logic circuits.

- a) More
- b) Less
- c) Equal
- d) Very High

Answer: b

Explanation: CMOS logic dissipates less power than NMOS logic circuits because CMOS dissipates power only when switching (“dynamic power”). Thus, CMOS has less power consumption and is more efficient.

9. Semiconductors are made of _____

- a) Ge and Si
- b) Si and Pb
- c) Ge and Pb
- d) Pb and Au

Answer: a

Explanation: Semiconductors are made of Silicon (Si) and Germanium (Ge).
Semiconductors are devices having conductivity between conductors and insulators.

10. Which chip were the first RTC and CMOS RAM chip to be used in early IBM computers, capable of storing a total of 64 bytes?

- a) The Samsung 146818
- b) The Samsung 146819
- c) The Motorola 146818
- d) The Motorola 146819

Answer: c

Explanation: The Motorola 146818 was the first RTC and CMOS RAM chip to be used in early IBM computers; capable of storing a total of 64 bytes.

8.) BiCMOS Logic Circuits

1. BiCMOS is an evolved semiconductor technology that integrates two formerly separate semiconductor technologies those of the _____

- a) CMOS and FET
- b) MOSFET and CMOS
- c) BJT and CMOS
- d) BJT and MOSFET

Answer: c

Explanation: BiCMOS is an evolved semiconductor technology that integrates two formerly separate semiconductor technologies those of the BJT and CMOS. CMOS ensures low power dissipation, smaller noise margin and higher packaging density, while BJT offers high switching speed and improved noise performance.

2. We use BJT in BiCMOS because of its _____

- a) High speed
- b) High gain
- c) Low output resistance
- d) All of the Mentioned

Answer: d

Explanation: Bipolar junction transistors offer high speed, high gain and low output resistance, which are excellent properties for high-frequency analog amplifiers. It also offers high switching speed and improved noise performance.

3. We use CMOS transistor in BiCMOS because of its _____

- a) High input resistance
- b) Simple construction
- c) Low power logic gates
- d) All of the Mentioned

Answer: d

Explanation: CMOS ensures low power dissipation, smaller noise margin and higher packaging density. It also offers high input resistance and is excellent for constructing simple, low-power logic gates.

4. When was BiCMOS technology introduced?

- a) 1990
- b) 1991
- c) 1992
- d) 1989

Answer: a

Explanation: BiCMOS technology was introduced in 1990s. BiCMOS is an evolved semiconductor technology that integrates two formerly separate semiconductor technologies those of the BJT and CMOS. CMOS ensures low power dissipation, smaller noise margin and higher packaging density, while BJT offers high switching speed and improved noise performance.

5. On which technology the Pentium is based?

- a) MOSFET
- b) CMOS
- c) BiCMOS
- d) BJT

Answer: c

Explanation: The Pentium technology is based on BiCMOS technology because BiCMOS circuits use the characteristics of each type of transistor most appropriately. CMOS ensures low power dissipation, smaller noise margin and higher packaging density, while BJT offers high switching speed and improved noise performance.

6. The advantage of BiCMOS circuit is that it gives _____

- a) Large fan-out
- b) Large fan-in
- c) Small fan-in
- d) Small fan-out

Answer: a

Explanation: Fan-out is the measure of the maximum number of inputs that a single gate output can feed. BiCMOS circuits are particularly useful for logic with large fan-out (large capacitive load).

7. How many members a BiCMOS family have?

- a) 5
- b) 6
- c) 7
- d) 8

Answer: a

Explanation: BiCMOS is a combination of both CMOS and BJT technology. The BiCMOS family has mainly 5 members and those are ABT logic, ALB logic, ALVT logic, BCT logic and LVT logic.

8. What is LVT logic?

- a) Lower threshold voltage
- b) Lower supply threshold voltage
- c) Lower supply voltage while retaining TTL logic levels
- d) All of the Mentioned

Answer: c

Explanation: LVT stands for Low Voltage BiCMOS Technology which is an integration of less power dissipation and less noise of CMOS along with the high switching speed and output drive of BJT. LVT logic is nothing but the lower supply voltage while retaining TTL logic levels.

9. What does ALVT mean?

- a) All lower supply voltage while retaining TTL logic levels

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- b) ABT lower supply voltage while retaining TTL logic levels
- c) Advanced version of LVT logic
- d) All version of LVT logic

Answer: c

Explanation: ALVT stands for Advanced Low Voltage BiCMOS Technology (ALVT). It is a faster version of LVT family. It is an integration of less power dissipation and less noise of CMOS along with the high switching speed and output drive of BJT.

10. The full form of PTL is _____

- a) Pull transistor logic
- b) Push transistor logic
- c) Pass transistor logic
- d) Post Transistor Logic

Answer: c

Explanation: The full form of PTL is “pass transistor logic”. It combines different logic families to design an integrated circuit. Thus, it reduces the number of transistors used by scraping off the unnecessary transistors.

11. CVSL means _____

- a) Complementary Voltage Switch Logic
- b) Cascode Voltage Switch Logic
- c) Complemented Voltage Switch Logic
- d) Cascade Voltage Switch Logic

Answer: c

Explanation: Cascode Voltage Switch Logic (CVSL) refers to a CMOS-type logic family which is designed for certain advantages. It requires mainly N-channel MOSFET transistors to implement the logic using true and complementary input signals and also needs two P-channel transistors at the top to pull one of the outputs high.

9.) Compatibility or Interfacing

1. Compatibility refers to _____

- a) The output of a circuit should match with the input of another circuit
- b) The output of a circuit should match with the input of the same circuit
- c) The input of a circuit should match with the output of another circuit
- d) The input of a circuit should match with the output of same circuit

Answer: a

Explanation: The output(s) of a circuit or a system should match the input(s) of another

circuit or system that has different electrical characteristics. This is referred to as compatibility.

2. The method of connecting a driving device to a loading device is known as _____

- a) Compatibility
- b) Interface
- c) Sourcing
- d) Sinking

Answer: b

Explanation: The method of connecting a driving device to a loading device is known as interface. The output(s) of a circuit or a system should match the input(s) of another circuit or system that has different electrical characteristics. This is referred to as compatibility.

3. The first CML logic was introduced by General Electric in _____

- a) 1960
- b) 1981
- c) 1961
- d) 1990

Answer: c

Explanation: CML stands for Current Mode Logic. The first CML logic was introduced by General Electric in 1961.

4. Commercial ECL families are not nearly as popular as TTL and CMOS mainly because they _____

- a) Produces too much noise
- b) Consume too much power
- c) Have high fan-in
- d) Have high fan-out

Answer: b

Explanation: Commercial ECL families are not nearly as popular as TTL and CMOS mainly because they consume too much power. CMOS consumes the least power while TTL provides high speed.

5. The key to propagation delay in bipolar logic family is to prevent the transistors in a gate from _____

- a) Fan-in
- b) Fan-out
- c) Saturation
- d) Cut-off

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Answer: c

Explanation: The key to propagation delay in bipolar logic family is to prevent the transistors in a gate from saturation. In Saturation mode, the transistor is in “ON” mode, where it seems like a short circuit between collector and emitter.

6. Schottky families prevent the saturating using _____

- a) Transistors
- b) Schottky transistors
- c) Diodes
- d) Schottky diodes

Answer: d

Explanation: Schottky families prevent the saturating using Schottky diodes across the base-collector junctions of transistors. In Saturation mode, the transistor is in “ON” mode, where it seems like a short circuit between collector and emitter.

7. The basic idea of basic CML circuit came from an _____

- a) Inverter
- b) Buffer
- c) Transistor
- d) Both inverter and buffer

Answer: d

Explanation: CML is Current Mode Logic in which data is transmitted at high speed of Mbps. Since this circuit has both inverting and non-inverting output. So, It behaves like an inverter/buffer.

8. The full form of MECL is _____

- a) Mono emitter coupled logic
- b) Motorola emitter coupled logic
- c) Motorola emitter capacitor logic
- d) Both mono emitter and motorola coupled logic

Answer: b

Explanation: The full form of MECL is Motorola emitter coupled logic. ECL is a high speed BJT. It uses BJT differential amplifier with single input and restricted current to avoid the transistor going into saturation and thus turning off operation.

9. Motorola has offered MECL circuits in _____ logic families.

- a) 3
- b) 4

- c) 5
- d) 6

Answer: c

Explanation: Motorola has offered MECL circuits in five logic families: MECL I, MECL II, MECL III, MECL 10000 (MECL 10K) and MECL 10H000 (MECL 10KH). The full form of MECL is Motorola emitter coupled logic. ECL is a high speed BJT. It uses BJT differential amplifier with single input and restricted current to avoid the transistor going into saturation and thus turning off operation.

10. The latest entrant to the ECL family is _____

- a) ECL 10K
- b) ECL 100K
- c) ECL 1000K
- d) ECL 10000K

Answer: b

Explanation: The latest entrant to the ECL family is ECL 100K, having 6-digit part numbers. This family offers functions, in general, different from those offered by 10K series. This family operates with a reduced power supply voltage -4.5 V, has shorter propagation delay of 0.75 ns, and transition time of 0.7 ns. However, the power consumption per gate is about 40 mW.

11. All input of NOR as low produces result as _____

- a) Low
- b) Mid
- c) High
- d) High Impedance

Answer: c

Explanation: According to the properties of NOR gate, if all the input of NOR as low produces a result as high. While if any input of NOR is high, then it produces low output.

12. In RTL NOR gate, the output is at logic 1 only when all the inputs are at _____

- a) logic 0
- b) logic 1
- c) +10v
- d) floating

Answer: a

Explanation: RTL NOR gate behaves as NOR gate and the output of NOR gate will be 1 only when all the inputs are at logic 0. The output of NOR will be 0 if any of the input is 1.